# JOHN LINNU KALAVALA

**FULL-STACK WEB SYSTEMS DEVELOPER**

**Location:** Kakinada, Andhra Pradesh, India | **University:** JNTU College of Engineering Kakinada (JNTUK) | **Degree:** B.Tech in EEE
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## PROFESSIONAL SUMMARY

Undergraduate **Electrical & Electronics Engineering (EEE)** student at **JNTU College of Engineering Kakinada (JNTUK)**, with demonstrated commercial experience engineering and deploying production-grade web applications, business automation engines, and document generation systems. Proven track record architecting software for **Thomas International Music Academy (TIMA)**, **KAPL Enterprises**, **Bharat Pro Lifts**, and **Effisol Energy Solutions**. Combines rigorous electrical and computer engineering principles with modern JavaScript/Node.js web architectures, real-time booking engines, and dynamic batch PDF/DOCX automation.

## EDUCATION

### Bachelor of Technology (B.Tech) in Electrical & Electronics Engineering (EEE)

**2024 - 2028 (Expected)**

**Jawaharlal Nehru Technological University (JNTUK), College of Engineering Kakinada**, Andhra Pradesh
*Focus Areas: Electrical Circuit Theory, Digital Electronics, Microprocessors & Embedded Systems, Data Structures & Algorithms, Database Systems, Web Application Development.*

## PRODUCTION SYSTEMS & COMMERCIAL CLIENT EXPERIENCE

### Thomas International Music Academy (TIMA) | Studio Operations & Slot Booking PWA

**2026 | Live Production**

***Technologies:*** *Node.js, SQLite, Progressive Web App (PWA), Service Workers, Web Push Notifications, REST APIs*

- **Full-Stack Architecture:** Built a standalone Node.js and SQLite Progressive Web App managing studio operations across **18 daily operating hours (5:00 AM - 11:00 PM)**.
- **Coach Operations Portal:** Developed coach dashboard with automated audio notifications for unmarked sessions, roster filters, and transactional SQLite database snapshot backups.
- **Interactive Profile Tools:** Implemented touch photo cropping (1.0x to 3.0x pan/zoom) for student profiles and parent self-service attendance tracking.
- **Real-Time Slot Engine:** Created dynamic make-up slot discovery and reservation engine supporting **9 instrument rooms**.

### KAPL Enterprises | Solar Quotation Generator & Document Generator Pro

**2025 - 2026 | Commercial Systems**

***Technologies:*** *JavaScript (ES6+), jsPDF, Docxtemplater, PizZip, HTML5/CSS3, Business Logic Engines*

- **Quotation Calculation Engine:** Built an automated solar proposal generator computing custom kW loads, GST tiers, and central subsidies, producing branded A4 PDF proposals in under 3 seconds.
- **Batch Document Automation:** Engineered **Solar Document Generator Pro** to batch-generate **6 official legal agreements** (Model Agreement, Net Metering, Grounding, Annexures, Completion Reports) from a single entry.
- **Commercial Impact:** Replaced manual drafting workflows with a one-click combined DOCX batch export, saving **10+ hours per week** in administrative preparation.

### Bharat Pro Lifts | Elevator Technical Quotation Generator

**2025 | Commercial System**

***Technologies:*** *JavaScript, jsPDF, Mathematical Calculation Matrices, HTML5/CSS3*

- **Engineering Calculation Suite:** Developed an interactive quotation tool computing passenger/freight load limits, shaft dimensions, and door opening configurations.
- **Dynamic Proposal Export:** Implemented real-time cost calculation matrices with instant client-branded PDF quotation downloads and multi-step form validation.

### Effisol Energy Solutions | Commercial Solar Quotation Management Platform

**2025 | Commercial System**

***Technologies:*** *JavaScript, Dynamic PDF Generation, Product Catalog Search, Responsive UI*

- **Catalog Search & Proposals:** Built end-to-end quotation management platform with searchable Tier-1 module inventory, automatic GST computation, and exportable commercial client proposals.

### TechSummit | Conference Registration & Digital QR Ticketing Platform

**2026 | Live Platform**

***Technologies:*** *JavaScript, Vector QR Code Engine, Digital Badge Generation, Responsive Web UI*

- **Event Ticketing Engine:** Built conference registration portal with multi-tier ticket selection (General, All-Access, VIP) and automated digital badge generation with scannable QR verification.

## TECHNICAL SKILLS & ENGINEERING COMPETENCIES

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|---|---|
| **Core Languages:** | JavaScript (ES6+/Modern), TypeScript, Python, HTML5, CSS3, SQL, C/C++ (Embedded) |
| **Backend & Systems:** | Node.js (Native architectures), Express.js, RESTful APIs, Progressive Web Apps (PWA), Service Workers, Web Push |
| **Document Automation:** | jsPDF, Docxtemplater, PizZip, html2canvas, Automated GST & Subsidy Calculators, Batch DOCX/PDF Export |
| **Frontend & Responsive UI:** | Modern Responsive UI/UX, CSS Grid/Flexbox, GSAP Animations, Canvas Graphics, Cross-Browser Optimization |
| **Engineering & Databases:** | SQLite (Transactions & Backups), PostgreSQL, Git & GitHub, Linux CLI, Electrical Circuits, Microcontrollers |